

Dispatch

Assignment 3 MVP Implementation & Demo Report | Group Rapid Route

What was implemented

The approved narrow vertical slice is implemented as a TypeScript + Vite + Three.js browser simulation. Zone density drives a gravity OD matrix and peak-hour trip generation. Vehicles use IDM-style car-following and junction right-of-way rules. At departure, A* compares current live link travel times against a shorter distance-based candidate; the first longer-but-faster proof route is retained and highlighted in yellow.

Plan vs. reality

Assignment 2 requirement	Assignment 3 evidence
Zone density + OD matrix + peak hour	Implemented. The skyline follows zone density; the OD heatmap, AM period, top OD pairs, trip rate, and generated-trip counters are visible.
A few hundred cars with safe following	Implemented. The stress configuration reaches 200+ cars; IDM spacing and traffic metrics are visible. Tests report 208 peak vehicles.
Junction right-of-way	Implemented. Compatible movements share a junction; conflicting movements wait, exit reservations prevent blocking, and tests check invariants.
Cost-based routing proof	Implemented. The panel compares a selected route live cost with a shorter candidate live cost. A proof state is retained when the longer route is faster, with link count, distance, cost difference, and yellow overlay.
Deferred scope	Emergency dispatch, lane changing, calibrated driver behavior, adaptive signals, GIS, and multi-modal transport remain final-roadmap items.

AI utility

AI tools accelerated the data structures, routing implementation, UI instrumentation, tests, and documentation. Human review was required to tune demand, validate route-cost semantics, and add a reproducible longer-but-faster evidence state. AI is development assistance only; no LLM or network API is used at runtime.

Demo and roadmap

The demo shows AM demand and the OD matrix; the 60x stress test with 150+ cars and zero gridlock removals; the safety/evidence checklist; and the route panel in PROOF state. The next final-submission layer is emergency incident spawning, nearest-available dispatch, live rerouting for emergency units, response-time metrics, and further visual polish.

Submission links to complete

Playable web build	[REPLACE WITH GITHUB PAGES / VERCEL URL]
Video	[REPLACE WITH YOUTUBE / VIMEO URL]
Source	[REPLACE WITH PUBLIC GITHUB REPOSITORY URL]
Local run	From sim/: npm install, npm test, then npm run dev -- --host 0.0.0.0.

Evidence files: see docs/ASSIGNMENT3_DEMO_SCRIPT.md for the recording script and screenshot checklist.